

Yixuan (Edward) Xiong

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PROFESSIONAL EXPERIENCE

Seattle Office of Emergency Management | Software Engineer Intern | Seattle, WA | 01/2026 - 04/2026

- Served on a 4-person team as technical architect for an M365 workflow automation platform processing 2,000+ annual requests; designed a relational schema with event-driven notifications, reducing end-to-end request processing by 83%.
- Built a multi-criteria instructor matching engine replacing manual spreadsheets; improved match accuracy from 60% to 80% and saved 2.5 hours per weekly cycle.
- Implemented a PII-compliant data pipeline filtering identifiable fields before publishing aggregated metrics to real-time Power BI dashboards.

Gavel and Quill | Software Engineer Intern | Seattle, WA | 05/2025 - 08/2025

- Built an offline-capable map platform on AWS EC2 for energy site planning; designed a PostGIS spatial database ingesting NOAA and Census datasets, served through Leaflet-rendered vector tile layers.
- Engineered a geospatial ETL pipeline processing 300+ MB of daily NOAA data; optimized PostGIS GiST indexes and server-side Vector Tile caching, reducing map query latency by 60%.
- Implemented row-level Optimistic Locking with version counters to handle concurrent data edits during field sync cycles.

Mandarin Oriental Hotel Group | Revenue Operations Engineer | Guangzhou, China | 02/2022 - 02/2023

- Built Python ETL pipelines aggregating booking data from 4 OTA platforms into a unified schema; automated daily report generation replacing 8 hours/week of manual Excel consolidation.
- Developed a regression-based demand forecasting model predicting peak booking periods from historical occupancy and seasonal patterns.

PROJECTS

ResponseMatch - LLM Emergency Response Matching System | 04/2026

Python, LangGraph, Claude API, ChromaDB, RAGAS, FastAPI

- Architected a multi-agent system automating volunteer matching across 40+ Seattle 311 categories, parsing heterogeneous request data into urgency, geo-location, and keyword schemas.
- Designed a dual-track validation layer and integrated RAG via ChromaDB; reduced false-positive matches by 30% and improved RAGAS Faithfulness from 0.62 to 0.84.

MaterniFlow - Multi-Agent Clinical Operations System | 01/2026

Python, FastAPI, GPT-4 API, PostgreSQL, Next.js

- Designed a ReAct-based orchestrator agent that decomposes multi-step clinical requests into tool calls using GPT-4 function calling; implemented 3 specialized tool agents.
- Built guardrail middleware for LLM-generated DB operations with SQL injection detection, schema validation, and surgical time-slot conflict resolution.
- Trained a logistic regression model on synthetic OB/GYN admission data to predict patient length-of-stay and integrated predictions into a FastAPI endpoint.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, TypeScript, JavaScript, C/C++, SQL, Spring Boot, FastAPI, React, Next.js, Node.js

Cloud & DevOps: AWS (EC2, Lambda), Azure (Functions, Gov Cloud), Docker, Kubernetes, Nginx, Git, Linux, GitHub CI/CD

Data & AI/ML: PostgreSQL, MySQL, Redis, PostGIS, Pandas, Scikit-learn, LangChain, LangGraph, ChromaDB, RAGAS

EDUCATION

Northeastern University | Seattle, WA | Master of Science in Computer Science | 09/2023 - 05/2026

Courses: Database Systems, Distributed Systems, Machine Learning, Cloud Computing, Object Oriented Design

Macau University of Science and Technology | Macau SAR, China | Bachelor of Science | 09/2017 - 06/2021